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STAFF REPORT FOR THE 2017 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS[1]

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Lao P.D.R. faces a high risk of debt distress, based on an assessment of public external debt.[2] The assessment of high risk is reinforced by vulnerabilities related to the existence of currency mismatches and private external debt. Public external debt distress indicators are less elevated than in the 2016 DSA due to the rebasing of GDP, but the debt service-to-revenue ratio breaches its respective threshold in 2019, and remains well above this mark for the full projection period. In addition, several threshold breaches under stress tests reinforce the importance of building buffers for adverse shocks. The present value (PV) of the total public debt-to-GDP ratio also breaches its benchmark. Given the high share of foreign-currency-denominated debt, a large and sudden exchange rate depreciation could raise the level of several indicators significantly, including private external debt, placing debt dynamics on an unsustainable path. Though revenues from large resource projects and U.S. dollar returns of the exporting sectors are expected to mitigate risks over the long term, external borrowing should remain on concessional terms to reduce the debt service burden. There remains an urgent need to recalibrate fiscal policy to rebuild fiscal buffers, adopt clear guidelines for the issuance of sovereign debt and guarantees to help contain debt related risks, and improve debt management, including by developing a comprehensive medium-term debt management strategy and regularly performing a debt sustainability analysis to inform borrowing decisions.

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[1] This DSA has been prepared by IMF and World Bank staff, in consultation with the Lao P.D.R. authorities.

[2] The low-income country debt sustainability framework (LIC DSF) recognizes that better policies and institutions allow countries to manage higher levels of debt, and thus the threshold levels for debt indicators are policy-dependent. In the LIC-DSF, the quality of a country's policies and institutions is measured by the World Bank's Country Policy and Institutional Assessment (CPIA) index and classified into three categories: strong, medium, and weak. Lao P.D.R.'s policies and institutions, as measured by the CPIA, averaged 3.29 over the past three years. Since its average CPIA index has been above 3.25, but less than 3.75 for three years in a row, Lao P.D.R.'s policy performance remains classified as medium according to the "Staff Guidance Note on the Application of the Joint Fund-Bank Debt Sustainability Framework for Low-income Countries" (http://www.imf.org/external/np/pp/eng/2013/110513.pdf). Therefore, the relevant indicative thresholds for this category are: 40 percent for the PV of debt-to-GDP ratio, 150 percent for the PV of debt-to-exports ratio, 250 percent for the PV of debt-to-revenue ratio, 20 percent for the debt service-to-exports ratio, and 20 percent for the debt service-to-revenue ratio. These thresholds are applicable to public and publicly guaranteed external debt.

BACKGROUND

1. The 2016 Debt Sustainability Analysis (DSA) classified Lao P.D.R.'s risk of debt distress as high, worsening from the 2014 DSA which assessed risk as moderate, but on the borderline of high risk.

2. This DSA keeps the risk of debt distress at high. The indicative debt distress thresholds remain unchanged from the 2016 DSA, since the classification of Lao P.D.R.'s policy performance, according to the Country Policy and Institutional Assessment (CPIA) index, remains moderate. Under the baseline scenario of the current DSA, one of the external and public debt distress indicators, the debt service-to-revenue ratio, breaches the policy-dependent indicative threshold in a substantial and prolonged fashion. All other indicators remain below their respective thresholds under the baseline, an improvement compared to the last DSA, mainly due to a rebasing of GDP in 2017. Overall, the net present value (PV) of external debt follows a downward trend. Under the alternative scenarios, some indicators breach the thresholds for some periods, and the strongest breaches occurred under a one-time depreciation shock.

Lao P.D.R.: External Public Debt Indicators

	Indicative thresholds	End-2016
Present value of debt, as a percent of:		
GDP	40	33.1
Exports	150	100.5
Revenue	250	221.7
Debt service, as a percent of:		
Exports	20	7.0
Revenue	20	15.5

Sources: Lao P.D.R. authorities; and IMF and World Bank estimates.

Lao P.D.R. Debt-to-GDP Ratio under Rebased GDP

	old GDP	rebased GDP	
	2015	2015	2016
Nominal GDP (in bill. USD)	12.6	14.4	15.9
Public sector debt (in percent of GDP)	65.8	57.7	58.5
PPG external debt (in percent of GDP)	51.7	45.4	46.6

Sources: Lao P.D.R. authorities; and IMF and World Bank staff estimates.

3. Lao P.D.R.'s external public and publicly guaranteed (PPG) debt has risen for the past few years. The nominal stock of PPG external debt increased from US$6.5 billion at end-2015 to about US$7.4 billion at end-2016, due mainly to higher borrowing from Thailand and China and sovereign bond issuance in the Thai market. The rise in debt was in part driven by heavy investment in power generation projects, part of the strategy to use the country's abundant hydropower resources to export energy to the rapidly growing neighborhood.[3] In addition, the government has continued issuing bonds on the Thai market to support the budget. The PPG external debt ratio increased from 45.4 percent of GDP at end-2015 to 46.6 percent of GDP at end-2016. The corresponding net present value (PV) of PPG external debt stands at 33.1 percent at-end 2016, well below the 40 percent indicative threshold.

Lao P.D.R.: Stock of External PPG Debt at End-2016

	In Billions of U.S. Dollars	As a Share of Total External Debt	In Percent of GDP
Total	7.4	100	46.6
Multilateral	1.6	21.1	9.8
Bilateral	4.7	63.7	29.7
Commercial 1/	1.1	15.1	7.1

Sources: Lao P.D.R. authorities; and IMF and World Bank staff estimates.
1/ Commercial debt includes Thai bond issuance.

[3] The installed capacity in Lao P.D.R.'s power system increased from around 600MW in early 2000 to above 6,000MW most recently with most of the generated electricity is exported to Thailand. The installed capacity is expected to reach above 10,000MW by 2020.

Box 1. Section of the Kunming – Singapore Railway Line

The project involves the construction of a 420-kilometer single track electrified rail line from Vientiane to the northern border with China. Around 60 percent of the railway line will go through tunnels or on bridges. The railway line is a section of the proposed Kunming – Singapore Trans Asian Railway corridor. Construction activities started in 2017 and are expected to be completed in 2021.

The Lao P.D.R. section project cost has been estimated at US$6.7 billion, of which 30 percent will be provided by a joint venture company formed between Lao P.D.R. and China. Lao P.D.R. will contribute 30 percent of the capital of this company (or around US$700 million) in annual installments over the medium term. Of this, US$480 million is borrowed from China while the remaining funds will be provided by the Budget. The Lao P.D.R. Ministry of Finance has noted that no sovereign guarantee will be provided. Per the 2012 Feasibility Study, the IRR is 4.56 percent and the repayment period of investment is 23 years.

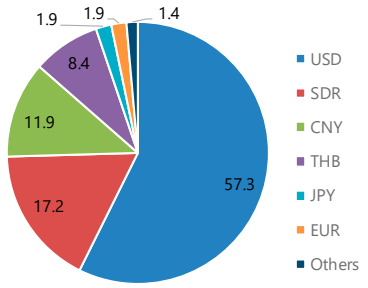
4. Bilateral creditors have been a greater source of loans than multilateral creditors in 2016, and this trend is expected to continue over the projection period. Bilateral creditors—mainly China, Thailand, Japan, and Korea—accounted for 63.7 percent of total external PPG debt at end-2016. Multilateral creditors consist mainly of the Asian Development Bank (ADB—11.1 percent of total external PPG debt), and the International Development Association (IDA—7.6 percent of total external PPG debt). Sovereign bonds have also been issued in the Thai capital market starting in May 2013. In 2016, bonds equaling around US$334 million were issued, and US$426 million in 2017. The outstanding sovereign bond debt at end-2016 was US$1,114 million, 15.1 percent of total external PPG debt. The issuance of bonds with different maturities indicates the authorities' goal to establish a yield curve. Most recently, the authorities could push the curve to 15 years. A credit rating agency based in Thailand (TRIS Rating Co., Ltd) provided an investment grade rating of BBB+ for the Lao government bonds in the Thai market. This rating is based on strong growth, Lao's abundance of natural resources, rising government revenue from hydropower, and the government's commitment to modernize the economy and alleviate poverty.

5. About 57 percent of total external PPG outstanding debt is contracted in U.S. dollars. The rest consists of SDR (17 percent), yuan (12 percent), baht (8 percent), euro, yen (each 2 percent) and others. The currency composition of the portfolio has changed slightly between end-2015 and end-2016, mostly due to possible issues in the recording of the original currency and eventual shifts in the currency of exposure.

6. The high, though declining, concessionality of official borrowing helps to reduce the external debt service burden. The PPG external debt service-to-exports ratio is expected to remain below the policy-dependent indicative threshold throughout the projection period in the baseline scenario. However, the PPG external debt service-to-revenue ratio is expected to exceed the threshold for most of the projection period, due to relatively weak tax revenues. Furthermore, given a high share of U.S. dollars in the currency composition of outstanding external debt and declining concessionality of new borrowing under the current DSA assumptions, these debt service ratios are sensitive to large sudden currency depreciation shocks.

7. Even though the PV of external debt-to-GDP ratio remains below its indicative threshold, the high debt service-to-revenue ratio and vulnerability to stress tests underscore the need to strengthen debt management capacity, including drawing up a comprehensive medium-term debt management strategy. When contracting new debt, debt sustainability considerations should be taken into account, particularly because the country is expected to shift gradually from concessional to more market-based terms. Additional near-term external borrowing, for example to finance large projects, could move the debt burden indicators again closer or over indicative thresholds, potentially undermining debt sustainability. A mitigating factor for Lao P.D.R.'s external debt burden lies in the prospective returns on the hydropower projects that have been financed in part by external PPG debt. The long-term power purchase agreements for these projects and the resulting government revenues in the form of royalties, dividends, and profit tax payments should help to reduce the risk of debt distress in the long run.

Currency Composition of External PPG Debt
(Percent of total)



Sources: Lao P.D.R. authorities; and IMF and World Bank staff estimates.

8. Recorded domestic PPG debt fell from 14.1 percent of GDP at end-2015 to about 11.9 percent of GDP at end-2016 on the back of higher GDP due to rebasing. Domestic debt consists of bond/T-bill holdings and the legacy debt of Bank of Lao P.D.R.'s direct lending to local government's off-budget infrastructure projects in the past. The resolution of domestic payment arrears related to public infrastructure projects is underway and is expected to add around 2 percent of GDP to domestic debt in 2018. Given higher costs of domestic borrowing, the share of domestic PPG debt remains relatively small. Going forward, as domestic financial markets deepen, the share of domestic public debt is likely to increase. Total domestic and external PPG debt stood at 58.5 percent of GDP at end-2016, compared to 57.7 percent at end-2015.

ASSUMPTIONS UNDERLYING THE DEBT SUSTAINABILITY ANALYSIS

9. The medium-term macroeconomic assumptions underlying the DSA are summarized in Box 2. The baseline scenario—which is based on current policies and consistent with the macroeconomic framework presented in the staff report—projects annual GDP growth to moderate to 6.8 percent in both 2017 and 2018. Average real GDP growth over the projected period (2017–37) is expected to be 6.4 percent, marginally higher than in the 2016 DSA, reflecting robust energy exports and a more favorable external environment, including higher growth in

Lao P.D.R. Macroeconomic Assumptions
Comparison with previous DSA
(Average over the 20 years projection period)

	2016 DSA	2017 DSA
GDP growth	6.3	6.4
GDP deflator in U.S. dollar terms (percent)	2.1	1.8
Non-interest current account deficit	11.3	7.3
Primary deficit	2.8	2.8

Sources: Lao P.D.R. authorities; and IMF and World Bank staff estimates.

China and a rebound in prices of key exports such as commodities and food. GDP deflator growth (in USD terms) is projected to be about 1.8 percent, slightly lower than in the 2016 DSA, in line with lower global inflation. The non-interest current account deficit is projected to fall to 7.3 percent of GDP due to higher electricity exports and rebased GDP. While, on the fiscal side, the primary deficit is expected to remain at its historical average of 2.8 percent of GDP.

Box 2. Baseline Scenario—Underlying Assumptions (2017–37)

Real GDP growth is projected to average 6.9 percent during 2017–22. Growth in 2017 at 6.8 percent reflects robust energy exports and a favorable external environment, including higher growth in China and a rebound in commodity prices. Real GDP is expected to moderate to 6.4 percent on average during 2017–37, as production in the resource sector reaches maturity but, partially, will be offset by strong electricity exports. Graduation from Least Developed Country (LDC) status is projected around the early-2020s.

Inflation (measured by GDP deflator in USD terms) is projected to average about 1.8 percent in 2017–37, slightly lower than in the 2016 DSA, in line with lower global inflation.

The balance of payments continues to be driven by developments in the resource sector, which has an important bearing both on the current account and the capital and financial accounts. The non-interest current account deficit is estimated to have narrowed to about 10.4 percent of GDP in 2017 from 18.1 percent during 2014, but is expected to widen over the medium term, as the railway project is implemented, before it declines to 7.3 percent of GDP on average in the longer term, as the resource balance improves due to the coming on-line of large-scale power projects. FDI inflows are assumed to be robust, driven by growing investment inflows into both resource and non-resource sectors.

External financing is assumed to remain largely on concessional terms in the near term. In the longer-run, however, the degree of concessional financing decreases with economic development, while the new disbursement schedule will shift from multilateral to commercial and bilateral creditors.

- **Multilateral Creditors**: Projected loan disbursements in the medium term are relatively higher than the authorities' projections. New disbursements from IDA are expected to register US$60-70 million a year. Over the longer term, the share of multilateral loans in total disbursements is expected to decline.
- **Bilateral and Commercial Creditors**: Over the medium and longer terms, project loan disbursements are expected to increase, as creditors provide support to the government's development agenda. As Lao P.D.R. exits from low income country status, a larger share of external borrowing is expected to come from bilateral and commercial creditors, with a lower degree of concessionality. This DSA incorporates historical and projected sovereign bond issuance in the Thai market and assumes their continuous roll-over and new bond issuances in the medium term.

Fiscal policy is projected to be neutral in the medium-term. The primary deficit is projected to peak at 3.7 percent of GDP in 2016 and is projected to slightly fall to 2.8 percent over the medium term. Over the long-term the average primary deficit is expected to remain at around 2.8 percent, as modest improvements in non-mining revenue collection come on line, while capital expenditure is expected to ease and other expenditure categories are viewed to remain constant as a percent of GDP.

Domestic debt is expected to increase over the long-term as the country relies more on domestic funding. Going forward, as global interest rates are projected to rise and domestic financial markets deepen, a larger share of financing needs is likely to be satisfied by domestic creditors.

10. A higher level of financing is assumed than in the 2016 DSA. To meet the country's financing needs, a higher level of new borrowing is projected to finance investment that would support the country's ambition to graduate from LDC status by 2020. External financing is assumed to remain partially on concessional terms, though with declining grant elements, in the near future, save for sovereign bond issuance on the Thai market. Going forward, however, the new disbursement schedule is assumed to rely less on multilateral creditors, and more on bilateral and commercial creditors. Multilateral assistance will slowly shift from concessional financing to credit-based conditions, and the ADB and the World Bank Group are expected to remain the principal suppliers of multilateral credit, with IDA slowly phasing out and being replaced by IBRD loans with less concessional terms. The level of grant financing is projected to decline over the projection period, adding to the higher projected borrowing needs. As the domestic financial market deepens, the private sector is assumed to rely more on domestic sources of financing, lowering the need for foreign borrowing in the long term.

DEBT SUSTAINABILITY

A. External Debt Sustainability Analysis

11. Under the baseline scenario, the three solvency indicators remain below their respective policy dependent indicative thresholds. The PV of external debt-to-GDP ratio, PV of external debt-to-exports ratio and the PV of external debt-to-revenue ratio are projected to remain below respective policy dependent indicative thresholds and decline in the medium term (Figure 1). The change from the previous DSA is driven primarily by higher GDP following rebasing.

12. The current DSA breaches the policy dependent indicative threshold for the debt service-to-revenue ratio. This breach is prolonged and substantial, providing grounds for the high-risk assessment. The other debt service indicator, the debt service-to-exports ratio, remains well below the policy-dependent indicative threshold during the entire forecast period under the baseline scenario. A mitigating factor for the breach in the debt service to revenue ratio is the expected diversification in the public revenue base in Lao P.D.R. due to returns on the hydropower projects that have been financed in part by the external PPG debt, and the long-term power purchase agreements for these projects in U.S. dollars.

13. Under the historical scenario, in which key variables are set at their 10-year historic average, debt dynamics become unsustainable for almost all debt indicators. The historical scenario for Lao P.D.R. builds in more adverse conditions for the current account deficit, real GDP growth, and growth of exports of goods and services than assumed under the baseline scenario, leading to higher debt accumulation rates and placing debt dynamics on an unsustainable path. As shown in Figure 1, under the historical scenario, all debt indicators except the debt service-to-exports ratio are projected to breach the respective policy dependent indicative thresholds.

14. Moreover, debt dynamics are markedly worse under stress test scenarios, with exchange rate depreciation risk having the largest impact. An abrupt exchange rate depreciation remains the most important risk to sustainability, given a large share of foreign currency debt and a very thin international reserves cushion. As shown in Figure 1, a one-off 30 percent depreciation shock would cause

the breach of the indicative threshold of the PV of debt-to-GDP ratio, the PV of debt-to-revenue ratio, and debt service-to-revenue ratio over a prolonged period.

B. Public Sector Debt Sustainability Analysis

15. The PV of public sector debt in percent of GDP is projected to breach the benchmark for several years and decline over the long run under the baseline scenario. Current public sector debt dynamics show an improved short-term situation compared to the 2016 DSA, but a worsening of long-run developments. The PV of public sector debt was estimated at 45.0 percent of GDP in 2016 and is expected to increase to around 65 percent before falling to around 54 percent in the long-term under the baseline scenario, below the public debt benchmark of 56 percent. Public sector debt is estimated at 58.5 percent at end-2016, compared to 57.7 percent of GDP at end-2015 in the previous DSA, and is expected to rise to 74.6 percent of GDP by 2026 before declining over the long run.

16. The PV of public sector debt remains sensitive to a large, abrupt exchange rate depreciation and the realization of contingent liabilities. Owing to significant reliance on external borrowing, a sudden 30-percent depreciation of the kip against the U.S. dollar would immediately raise the PV of public sector debt-to-GDP in the medium-term, with unfavorable implications for debt sustainability. Also, given the fragile public banks, recapitalization costs for three banks, which are estimated to be at least about US$250 million (1.8 percent of GDP), could add to the debt burden. Finally, while growing electricity exports are expected to mitigate the risks of public debt sustainability, these will only materialize if markets are secured for the rapidly expanding production at above-cost recovery prices.

17. The above threshold levels and susceptibility to shocks under stress test scenarios highlight risks and the importance of fiscal consolidation over the medium term. As shown in Figure 2 for the fixed primary balance scenario, which assumes an unchanged primary balance from 2017 for the entire projection period, the PV of the debt-to-GDP, debt-to-revenue, and debt service-to-GDP ratios would be higher than the baseline over the projection period.

AUTHORITIES' VIEWS

18. The authorities broadly agree with the overall assessment, recognizing the need to take further measures to reduce the debt burden. The government has taken steps to limit its borrowing and limit the contracting of additional debt to concessional borrowing. The government has taken measures to maintain its fiscal deficit and economize budget expenditure, but sees the need for growth supporting spending measures and increased social sector spending; implementing further measures and new legal rules takes time. A new Procurement Law has been implemented, the new Debt Management Law is being prepared, and the Ministry of Finance is in the process of reorganization to merge the management of all debt (domestic and foreign) in one department.

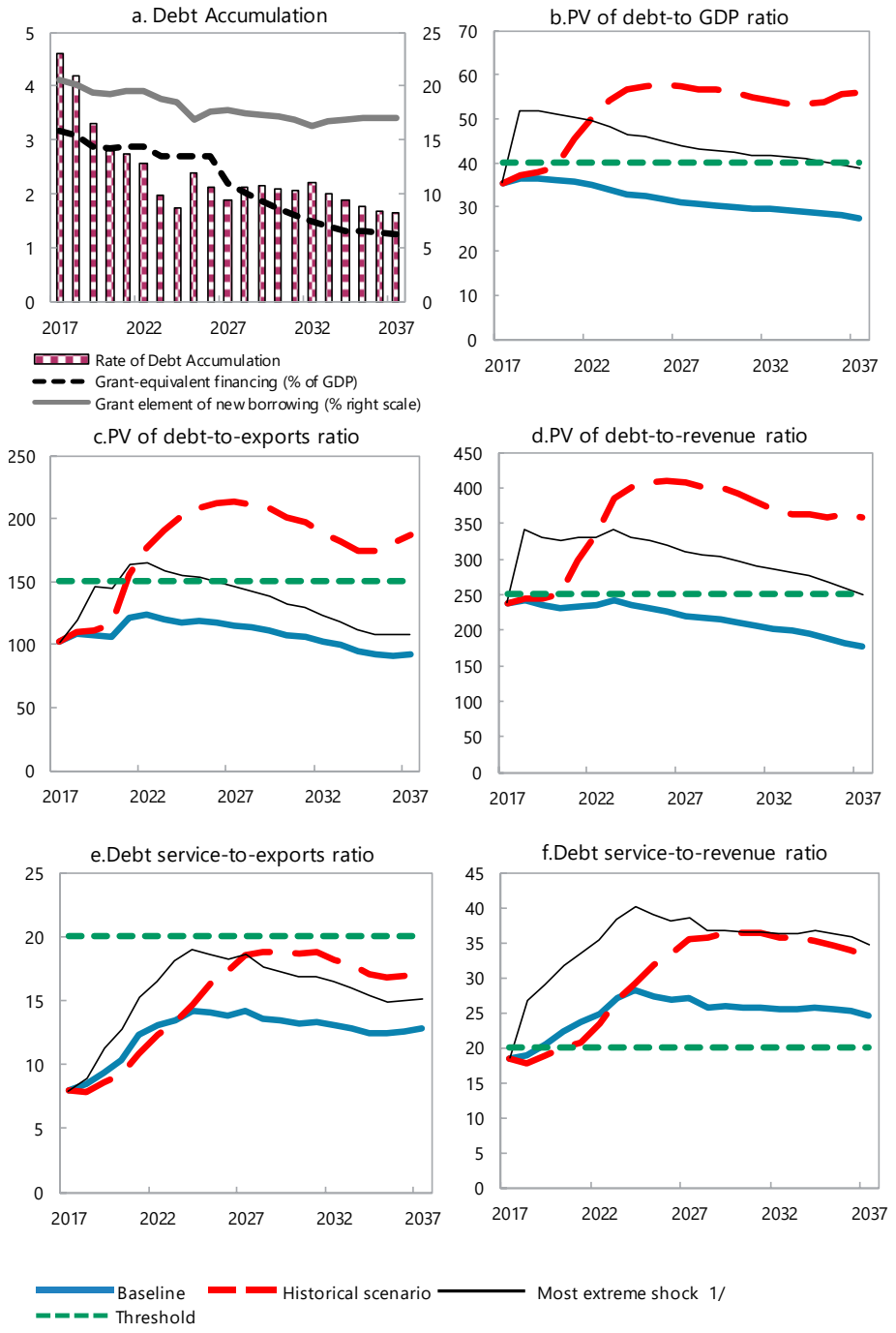
19. The authorities highlight that a significant part of the external debt is related to large, commercially viable hydroelectric projects and do not foresee difficulties in servicing debt. They project that energy projects will generate high and stable economic returns upon completion and will

supply enough foreign exchange to service debt. A relatively long maturity profile of loans, as well as U.S. dollar returns of the exporting sectors, would help mitigate the risks of debt distress.

CONCLUSION

20. Lao P.D.R.'s risk of external debt distress remains high, suggesting the urgent need to tighten fiscal policy, strengthen public financial management, and develop a comprehensive medium-term debt management strategy. The PV of the debt service-to-revenue ratio breaches the respective policy-dependent indicative thresholds in the external DSA, and the PV of debt-to-GDP ratio exceeds its benchmark in the public DSA. Both external and public debt indicators are susceptible to shocks, particularly a sudden depreciation of the kip/U.S. dollar exchange rate. Currency mismatches on bank balance sheet and the high level of private external debt reinforce these vulnerabilities. The higher debt stock at end-2016 undermines fiscal space for countercyclical needs and potential banking sector or other contingent costs. Given the considerable share of foreign currency denominated debt, a large sudden exchange rate depreciation could significantly raise the level of those indicators, putting debt dynamics on an unsustainable path. To reduce the debt burden, external borrowing should be contracted on concessional terms as much as possible. The authorities should recalibrate fiscal policy to rebuild fiscal buffers through stronger revenue mobilization efforts and expenditure rationalization, adopt clear guidelines for the issuance of sovereign debt and guarantees, and accelerate the strengthening of the debt management functions including developing a comprehensive medium-term debt management strategy and a regular debt sustainability analysis to inform borrowing decisions.

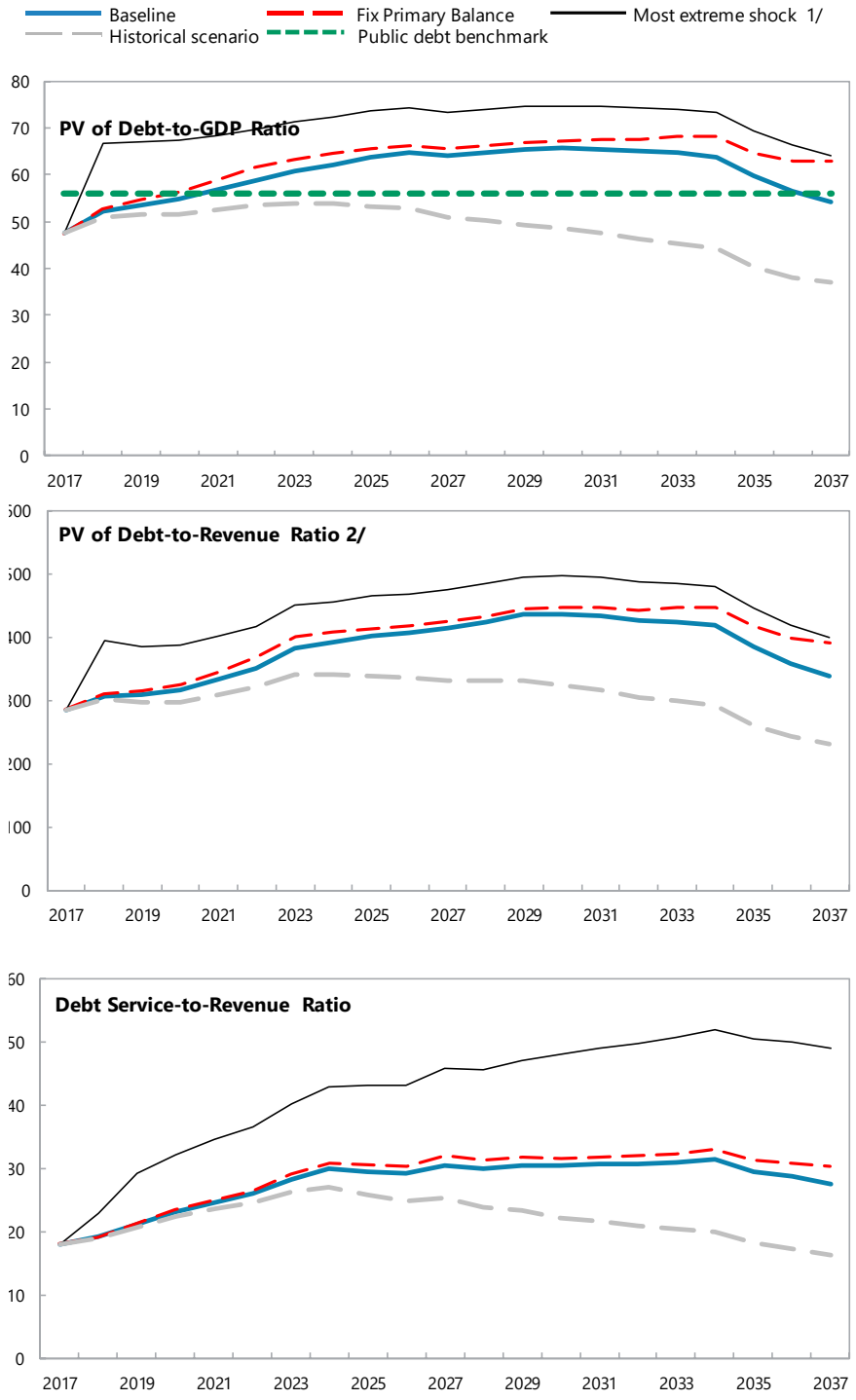
Figure 1. Lao P.D.R.: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2017–37 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2027. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2. Lao P.D.R.: Indicators of Public Debt Under Alternative Scenarios, 2017–37 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2027.
2/ Revenues are defined inclusive of grants.

Table 1. Lao P.D.R.: External Debt Sustainability Framework, Baseline Scenario, 2014–37 1/

(Percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2014	2015	2016			2017	2018	2019	2020	2021	2022	2017-2022 Average	2027	2037	2023-2037 Average
External debt (nominal) 1/	**100.1**	**102.7**	**104.5**			**113.7**	**119.8**	**122.1**	**120.5**	**113.8**	**107.8**		**82.8**	**57.2**	
of which: public and publicly guaranteed (PPG)	45.6	45.4	46.6			49.1	49.9	48.8	47.4	46.9	46.5		40.7	35.1	
Change in external debt	9.4	2.6	1.8			9.2	6.1	2.3	-1.6	-6.7	-6.0		-3.9	-1.9	
Identified net debt-creating flows	4.0	0.8	-5.1			0.0	1.9	1.0	-0.1	-6.7	-5.0		-1.1	-1.3	
Non-interest current account deficit	**18.1**	**16.7**	**9.6**	**16.5**	**5.7**	**10.4**	**12.5**	**11.8**	**11.0**	**9.1**	**8.3**		**6.8**	**4.1**	6.0
Deficit in balance of goods and services	23.7	22.1	11.5			12.3	14.3	12.8	11.6	10.8	9.9		7.9	5.6	
Exports	38.3	32.1	32.9			34.8	33.8	34.0	33.9	29.3	28.4		26.9	30.0	
Imports	62.0	54.2	44.4			47.2	48.2	46.8	45.5	40.1	38.3		34.8	35.6	
Net current transfers (negative = inflow)	-5.9	-5.6	-2.6	-3.0	1.5	-2.7	-2.7	-2.6	-2.6	-2.5	-2.5		-2.1	-1.2	-1.7
of which: official	-4.5	-4.3	-1.3			-1.5	-1.5	-1.5	-1.5	-1.5	-1.5		-1.3	-0.4	
Other current account flows (negative = net inflow)	0.3	0.2	0.7			0.7	0.9	1.6	1.9	0.8	1.0		1.1	-0.3	
Net FDI (negative = inflow)	**-7.2**	**-9.5**	**-7.0**	**-7.4**	**2.3**	**-6.2**	**-5.8**	**-5.0**	**-5.0**	**-10.3**	**-8.5**		**-4.6**	**-3.2**	-4.3
Endogenous debt dynamics 2/	**-6.9**	**-6.4**	**-7.7**			**-4.1**	**-4.8**	**-5.8**	**-6.1**	**-5.5**	**-4.9**		**-3.4**	**-2.2**	
Contribution from nominal interest rate	1.9	1.3	2.3			2.6	2.3	1.9	1.7	2.2	2.3		1.9	1.0	
Contribution from real GDP growth	-6.2	-6.7	-6.5			-6.7	-7.2	-7.7	-7.8	-7.6	-7.1		-5.3	-3.1	
Contribution from price and exchange rate changes	-2.6	-0.9	-3.5			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**5.3**	**1.8**	**6.9**			**9.1**	**4.2**	**1.3**	**-1.4**	**0.0**	**-1.0**		**-2.8**	**-0.7**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	...	
PV of external debt 4/	...	...	91.0			100.1	106.5	109.7	109.1	102.5	96.4		73.2	49.6	
In percent of exports	...	...	276.7			287.3	314.9	322.6	322.0	350.2	339.4		272.0	165.7	
PV of PPG external debt	**...**	**...**	**33.1**			**35.5**	**36.6**	**36.5**	**36.0**	**35.6**	**35.1**		**31.0**	**27.5**	
In percent of exports	**...**	**...**	**100.5**			**101.8**	**108.2**	**107.4**	**106.3**	**121.6**	**123.4**		**115.3**	**91.9**	
In percent of government revenues	**...**	**...**	**221.7**			**237.6**	**241.7**	**234.0**	**230.6**	**233.0**	**234.3**		**219.5**	**176.0**	
Debt service-to-exports ratio (in percent)	**13.8**	**13.7**	**21.8**			**22.9**	**23.3**	**24.3**	**24.6**	**28.9**	**29.0**		**32.0**	**15.4**	
PPG debt service-to-exports ratio (in percent)	**6.0**	**5.6**	**7.0**			**7.9**	**8.4**	**9.4**	**10.3**	**12.3**	**13.1**		**14.2**	**12.7**	
PPG debt service-to-revenue ratio (in percent)	**14.2**	**11.0**	**15.5**			**18.5**	**18.8**	**20.4**	**22.3**	**23.6**	**24.8**		**27.0**	**24.3**	
Total gross financing need (Billions of U.S. dollars)	2.1	1.7	1.6			2.1	2.7	3.0	3.1	1.7	2.1		4.3	4.7	
Non-interest current account deficit that stabilizes debt ratio	8.7	14.1	7.8			1.2	6.4	9.6	12.5	15.8	14.3		10.7	6.1	
Key macroeconomic assumptions															
Real GDP growth (in percent)	7.6	7.3	7.0	7.7	0.3	6.8	6.8	7.0	7.0	6.9	6.8	6.9	6.6	5.7	6.2
GDP deflator in US dollar terms (change in percent)	3.0	0.9	3.5	7.0	5.1	-0.1	1.1	2.2	2.2	1.9	1.9	1.5	1.9	1.9	1.9
Effective interest rate (percent) 5/	2.4	1.4	2.5	1.9	0.4	2.7	2.2	1.7	1.6	2.0	2.2	2.1	2.3	1.8	2.1
Growth of exports of G&S (US dollar terms, in percent)	18.9	-9.4	13.6	14.9	15.8	13.0	4.8	10.0	8.9	-5.8	5.7	6.1	7.2	5.2	8.6
Growth of imports of G&S (US dollar terms, in percent)	6.9	-5.5	-9.2	16.4	16.6	13.4	10.2	6.3	6.3	-3.9	3.9	6.0	8.1	8.6	7.7
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	20.6	20.1	19.3	19.2	19.6	19.6	19.7	17.8	17.1	17.3
Government revenues (excluding grants, in percent of GDP)	16.2	16.3	14.9			14.9	15.2	15.6	15.6	15.3	15.0		14.1	15.6	14.5
Aid flows (in Billions of US dollars) 7/	0.6	0.5	0.2			0.5	0.5	0.5	0.6	0.6	0.7		0.8	0.7	
of which: Grants	0.6	0.5	0.2			0.3	0.3	0.4	0.4	0.4	0.5		0.5	0.4	
of which: Concessional loans	0.0	0.0	0.0			0.2	0.2	0.2	0.2	0.2	0.2		0.3	0.3	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			3.2	3.1	2.9	2.8	2.9	2.9		2.2	1.3	1.9
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			36.9	37.6	38.9	39.2	39.1	39.3		34.9	23.8	30.5
Memorandum items:															
Nominal GDP (Billions of US dollars)	13.3	14.4	15.9			17.0	18.3	20.1	21.9	23.9	26.0		39.5	85.1	
Nominal dollar GDP growth	10.8	8.3	10.8			6.7	8.0	9.4	9.4	9.0	8.9	8.5	8.6	7.7	8.2
PV of PPG external debt (in Billions of US dollars)			5.2			6.0	6.7	7.3	7.9	8.5	9.1		12.2	23.3	
(PVt-PVt-1)/GDPt-1 (in percent)						4.6	4.2	3.3	2.8	2.8	2.6	3.4	1.9	1.7	2.0
Gross workers' remittances (Billions of US dollars)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	33.1			35.5	36.6	36.5	36.0	35.6	35.1		31.0	27.5	
PV of PPG external debt (in percent of exports + remittances)	...	...	100.5			101.8	108.2	107.4	106.3	121.6	123.4		115.3	91.9	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	7.0			7.9	8.4	9.4	10.3	12.3	13.1		14.2	12.7	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Lao P.D.R.: Public Sector Debt Sustainability Framework, Baseline Scenario, 2014–37

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate			Projections						
	2014	2015	2016	Average 5/	Standard Deviation 5/	2017	2018	2019	2020	2021	2022	2017-22 Average	2027	2037	2023-37 Average
Public sector debt 1/	57.9	57.7	58.5			61.1	65.3	65.9	66.2	67.9	70.1		73.6	61.7	
of which: foreign-currency denominated	45.6	45.4	46.6			49.1	49.9	48.8	47.4	46.9	46.5		40.7	35.1	
									0.0	0.0	0.0		0.0	0.0	
Change in public sector debt	1.7	-0.2	0.8			2.6	4.3	0.6	0.2	1.8	2.2		-1.0	-2.5	
Identified debt-creating flows	-2.9	-1.9	-0.6			1.2	-0.5	-1.7	-1.6	-1.1	-0.9		-0.9	-2.8	
Primary deficit	1.5	1.6	3.7	1.9	1.1	3.4	2.8	2.8	2.8	2.6	2.6	2.8	3.5	0.6	2.8
Revenue and grants	20.8	19.9	16.4			16.7	17.0	17.4	17.4	17.1	16.8		15.4	16.1	
of which: grants	4.6	3.6	1.5			1.8	1.8	1.8	1.8	1.8	1.8		1.3	0.4	
Primary (noninterest) expenditure	22.3	21.4	20.1			20.1	19.7	20.2	20.2	19.6	19.3		18.9	16.6	
Automatic debt dynamics	-4.4	-3.5	-4.3			-2.3	-3.3	-4.4	-4.5	-3.7	-3.5		-4.3	-3.4	
Contribution from interest rate/growth differential	-3.1	-3.5	-3.8			-3.3	-3.6	-4.4	-4.4	-3.8	-3.6		-4.4	-3.4	
of which: contribution from average real interest rate	0.9	0.4	0.0			0.4	0.3	-0.1	-0.1	0.5	0.8		0.2	0.1	
of which: contribution from real GDP growth	-4.0	-3.9	-3.8			-3.7	-3.9	-4.3	-4.3	-4.3	-4.3		-4.6	-3.5	
Contribution from real exchange rate depreciation	-1.3	0.0	-0.5			1.1	0.2	0.0	0.0	0.1	0.1		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	4.6	1.7	1.4			1.4	4.8	2.3	1.9	2.9	3.1		-0.1	0.3	
Other Sustainability Indicators															
PV of public sector debt	...	...	45.0			47.4	52.1	53.6	54.8	56.6	58.7		64.0	54.1	
of which: foreign-currency denominated	...	...	33.1			35.5	36.6	36.5	36.0	35.6	35.1		31.0	27.5	
of which: external	...	...	33.1			35.5	36.6	36.5	36.0	35.6	35.1		31.0	27.5	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	7.8	7.8	10.0			10.1	9.9	11.1	11.9	12.3	13.1		17.4	9.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	274.3			284.3	307.1	308.0	314.7	332.0	350.3		414.0	336.8	
PV of public sector debt-to-revenue ratio (in percent)	...	...	301.8			318.0	343.6	343.5	350.7	370.8	392.1		452.4	346.1	
of which: external 3/	...	...	221.7			237.6	241.7	234.0	230.6	233.0	234.3		219.5	176.0	
Debt service-to-revenue and grants ratio (in percent) 4/	14.3	12.1	15.3			18.1	19.1	21.1	23.1	24.6	25.9		30.5	27.5	
Debt service-to-revenue ratio (in percent) 4/	18.4	14.8	16.8			20.2	21.4	23.6	25.7	27.4	29.0		33.3	28.3	
Primary deficit that stabilizes the debt-to-GDP ratio	-0.1	1.8	2.9			0.9	-1.5	2.2	2.6	0.8	0.4		4.5	3.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.6	7.3	7.0	7.7	0.3	6.8	6.8	7.0	7.0	6.9	6.8	6.9	6.6	5.7	6.2
Average nominal interest rate on forex debt (in percent)	3.4	1.9	2.1	1.6	0.8	2.6	2.4	1.6	1.7	3.1	3.7	2.5	2.6	2.8	2.7
Average real interest rate on domestic debt (in percent)	1.7	3.1	-1.2	-1.1	3.1	1.0	1.3	0.4	0.2	0.4	0.3	0.6	-0.1	-0.7	-0.2
Real exchange rate depreciation (in percent, + indicates depreciation)	-3.1	0.1	-1.1	-4.3	4.4	2.5	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	5.7	2.3	3.0	4.5	2.7	1.4	2.2	3.0	3.2	3.0	3.0	2.6	3.0	3.0	3.0
Growth of real primary spending (deflated by GDP deflator, in percent)	-2.3	3.0	0.5	0.2	1.3	6.7	4.9	9.3	7.4	3.7	5.3	6.2	4.9	1.7	5.2
Grant element of new external borrowing (in percent)	...	...	...	...	...	20.6	20.1	19.3	19.2	19.6	19.6	19.7	17.8	17.1	...

Sources: Country authorities; and staff estimates and projections.

1/ Public sector debt statistics cover the central government gross debt.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 3. Lao P.D.R.: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017–37

(Percent)

	Projections							
	2017	2018	2019	2020	2021	2022	**2027**	2037
PV of debt-to GDP ratio								
Baseline	35	37	37	36	36	35	**31**	28
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	35	37	38	39	46	50	**58**	56
A2. New public sector loans on less favorable terms in 2017-2037 2/	35	38	39	39	40	40	**40**	44
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	35	36	36	36	35	35	**31**	27
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	35	38	42	42	41	40	**33**	28
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	35	36	36	36	35	35	**31**	27
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	35	38	39	38	38	37	**32**	27
B5. Combination of B1-B4 using one-half standard deviation shocks	35	34	32	32	31	31	**28**	26
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	35	52	52	51	50	50	**44**	39
PV of debt-to-exports ratio								
Baseline	102	108	107	106	122	123	**115**	92
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	102	109	112	116	156	177	**214**	187
A2. New public sector loans on less favorable terms in 2017-2037 2/	102	112	114	116	136	142	**150**	146
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	102	108	107	106	121	123	**115**	91
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	102	119	146	144	164	165	**146**	108
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	102	108	107	106	121	123	**115**	91
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	102	112	114	112	128	130	**118**	92
B5. Combination of B1-B4 using one-half standard deviation shocks	102	103	102	101	115	117	**112**	92
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	102	108	107	106	121	123	**115**	91
PV of debt-to-revenue ratio								
Baseline	238	242	234	231	233	234	**220**	176
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	238	244	243	251	299	336	**407**	358
A2. New public sector loans on less favorable terms in 2017-2037 2/	238	250	249	252	261	270	**286**	280
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	238	240	231	228	230	231	**216**	174
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	238	252	271	266	268	267	**236**	176
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	238	239	232	228	231	232	**217**	174
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	238	251	248	244	246	246	**225**	175
B5. Combination of B1-B4 using one-half standard deviation shocks	238	225	206	203	206	207	**199**	164
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	238	342	331	326	329	331	**310**	249

Table 3. Lao P.D.R.: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017–37 (concluded)

(Percent)

Debt service-to-exports ratio								
Baseline	8	8	9	10	12	13	**14**	13
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	8	8	9	9	11	12	**19**	17
A2. New public sector loans on less favorable terms in 2017-2037 2/	8	8	8	8	9	11	**14**	19
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	8	8	9	10	12	13	**14**	13
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	8	9	11	13	15	17	**19**	15
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	8	8	9	10	12	13	**14**	13
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	8	8	10	11	13	14	**15**	13
B5. Combination of B1-B4 using one-half standard deviation shocks	8	8	9	10	12	13	**14**	13
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	8	8	9	10	12	13	**14**	13
Debt service-to-revenue ratio								
Baseline	18	19	20	22	24	25	**27**	24
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	18	18	19	20	21	23	**35**	33
A2. New public sector loans on less favorable terms in 2017-2037 2/B70	18	19	17	18	18	21	**27**	36
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	18	19	20	22	23	25	**27**	24
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	18	19	21	24	25	27	**30**	25
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	18	19	20	22	24	25	**27**	24
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	18	19	21	23	24	26	**28**	24
B5. Combination of B1-B4 using one-half standard deviation shocks	18	18	19	21	22	22	**24**	23
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	18	27	29	32	34	35	**38**	35
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	12	12	12	12	12	12	**12**	12

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 4. Lao P.D.R.: Sensitivity Analysis for Key Indicators of Public Debt 2017–2037

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of Debt-to-GDP Ratio								
Baseline	47	52	54	55	57	59	64	54
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	47	51	51	51	52	54	51	37
A2. Primary balance is unchanged from 2017	47	53	55	56	59	62	65	63
A3. Permanently lower GDP growth 1/	47	52	54	55	57	59	65	56
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	47	52	53	54	56	58	63	52
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	47	52	54	55	57	59	64	54
B3. Combination of B1-B2 using one half standard deviation shocks	47	51	52	54	55	57	62	51
B4. One-time 30 percent real depreciation in 2018	47	67	67	67	68	70	73	64
B5. 10 percent of GDP increase in other debt-creating flows in 2018	47	61	62	63	64	66	70	58
PV of Debt-to-Revenue Ratio 2/								
Baseline	284	307	308	315	332	350	414	337
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	284	300	296	296	308	321	330	231
A2. Primary balance is unchanged from 2017	284	310	314	324	345	367	424	391
A3. Permanently lower GDP growth 1/	284	307	309	316	333	352	420	351
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	284	305	305	311	328	345	406	326
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	284	308	310	317	334	352	416	338
B3. Combination of B1-B2 using one half standard deviation shocks	284	304	302	308	324	342	401	320
B4. One-time 30 percent real depreciation in 2018	284	393	385	386	401	416	474	398
B5. 10 percent of GDP increase in other debt-creating flows in 2018	284	359	356	361	377	394	451	360
Debt Service-to-Revenue Ratio 2/								
Baseline	18	19	21	23	25	26	30	28
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	18	19	21	22	23	25	25	16
A2. Primary balance is unchanged from 2017	18	19	21	23	25	26	32	30
A3. Permanently lower GDP growth 1/	18	19	21	23	25	26	31	29
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	18	19	21	23	24	26	30	27
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	18	19	21	23	25	26	31	28
B3. Combination of B1-B2 using one half standard deviation shocks	18	19	21	23	24	25	30	26
B4. One-time 30 percent real depreciation in 2018	18	23	29	32	34	36	46	49
B5. 10 percent of GDP increase in other debt-creating flows in 2018	18	19	23	25	26	28	35	30

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.